# The Precambrian–Cambrian boundary problem: magnetostratigraphy of the Amadeus Basin, Central Australia

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*Summary.* A magnetostratigraphic study of Upper Proterozoic to early Cambrian sediments at 4 localities in Central Australia reveals the presence of a magnetic polarity pattern composed of 2 long quiet intervals, first of reversed then of normal polarity, which is then succeeded by a long mixed interval. Lithologic comparisons suggest that this pattern is a reflection of the behaviour of the geomagnetic field at that time, and probably is not a result of varying sedimentation rates. Two sampled sections in the Arumbera Sandstone separated by 80 km both contain the long normal interval followed by the interval of mixed polarity, the detailed comparison of which demonstrates that the Arumbera-Box Hole boundary is unconformable. Elements from the late Precambrian Mt Skinner fauna in the Arumbera Sandstone indicate that the long normal interval is entirely pre-Cambrian, and that the base of the Cambrian occurs within the interval of mixed polarity.

A comparison with an additional site sampled across a fossil occurrence in the Central Mount Stuart Formation at Central Mount Stuart indicates that at least 1 of the Upper Proterozoic fossils, *Hallidaya brueri*, has a considerable vertical distribution in time.

If used properly, the magnetic polarity pattern discussed herein offers the possibility of precise inter-continental correlations of stratigraphic sequences near the base of the Cambrian. It is suggested that a stratotype sequence for the base of the Cambrian be chosen in such a manner that the boundary position corresponds to a geomagnetic reversal.

## 1. Introduction

The recent application of superconducting magnetometers to the science of palaeomagnetism has produced a small quantum revolution within this field of study. It is now possible to measure and process thousands of palaeomagnetic samples of virtually any lithology in a matter of weeks, where formerly years of work were required. As a result, it is now possible to use the palaeomagnetic polarity information, the 'sign' of the magnetization, in sedimentary rocks as a stratigraphic tool for correlation between sequences of similar age. This necessarily depends on accepting the basic assumptions of palaeomagnetism concerning the behaviour of the geomagnetic field. These assumptions have been reviewed often, and will not be discussed here.

The aim of this present study is to examine in detail the magnetic polarity sequence recorded in fossil-bearing rocks on either side of the Precambrian–Cambrian boundary in Central Australia, in an effort to resolve a polarity pattern which could ultimately be used to unscramble the evolutionary events which occurred at this time. Much philosophical discussion has been given to the problem of defining the base of the Cambrian, and one proposed criterion has been the onset of biomineralization (Cloud, 1973; Cowie & Glaessner, 1975). Magnetic stratigraphy can potentially answer the question of whether or not this onset was a synchronous, world-wide event. It is therefore important to consider magnetostratigraphic information when examining potential stratotype sequences for the base of the Cambrian.

## 2. Geology

Three of the four localities studied in Central Australia (Fig. 1) are located in the northeastern portion of the Amadeus Basin, above the early Cambrian tectonic 'hinge line' described by Wells *et al.* (1970). The disruption of sedimentation resulting from the late Precambrian to early Cambrian Petermann Ranges Orogeny decreases rapidly N of this line, and is present, but barely perceptible, at the Valley Dam and Ross River localities. At Areyonga, an angular unconformity

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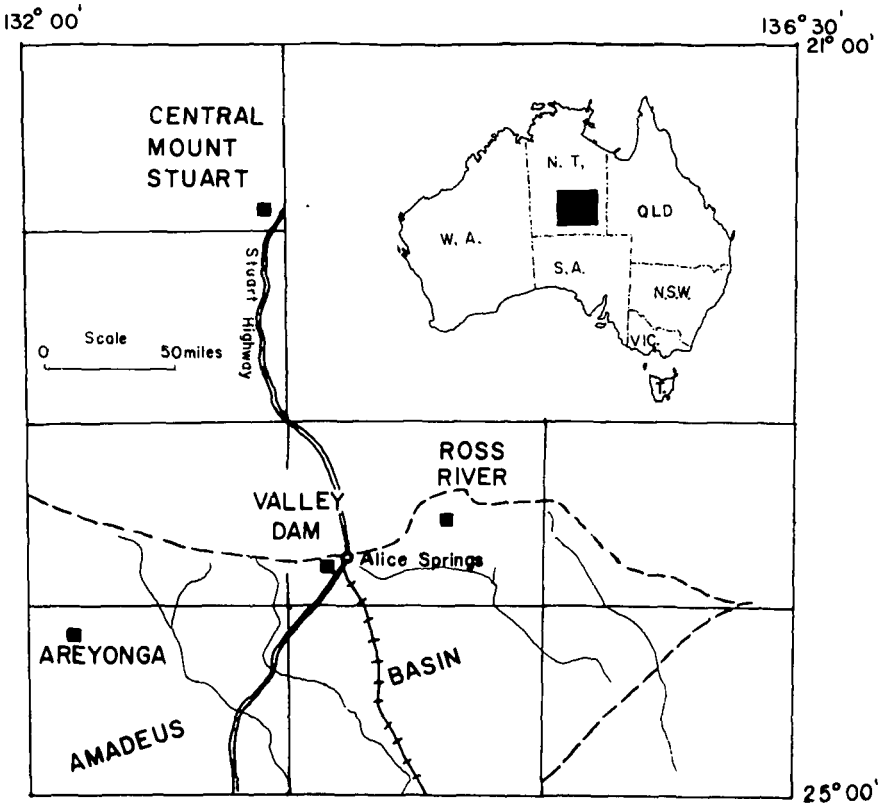



Figure 1. Index map showing location of measured magnetostratigraphic sections. The dashed line marks the approximate erosional boundary of the Amadeus Basin.

separates red and green claystones of the Pertatataka Formation from the Eninta Sandstone, which is a lateral equivalent of the Box Hole and Allua Formations sampled elsewhere. Late Palaeozoic deformation, including uplift, nappe formation, and regional folding, occurred during the Alice Springs Orogeny and is in large part responsible for the present erosional outline of the basin. No trace of metamorphism is present in any of the three localities sampled; however, a number of K–Ar and Rb–Sr systems have been reset during the orogeny.

Within the Amadeus basin, marine carbonates of Upper Proterozoic and Lower Cambrian age are more prevalent in the E, with clastics generally increasing towards the inferred source areas to the W and S. Consequently, the Ross River locality in the eastern part of the basin has abundant carbonates and claystones – these are absent in the equivalent units at Valley Dam and Areyonga further to the W. The magnetostratigraphic sampling of this study therefore includes a wide variety of facies of similar age.

The late Precambrian to early Cambrian fossil occurrences within the Amadeus Basin have been reviewed in detail by Daily (1974). Within the Arumbera Sandstone, an element of the late Precambrian Ediacara fauna has been found on float in the basal units near Deep Well, and a fossil assemblage from the late Precambrian Mt Skinner fauna has been found near the top of the formation at Valley Dam. A disconformity separates the Arumbera Sandstone from the Box Hole Formation, which contains early Cambrian trace fossils, such as *Rusophycus*. The Allua Formation, which rests conformably on the Box Hole, contains forms of *Diplocraterion* and *Plagiogmus*. Archaeocyathids, the first shelly fossils, appear in the lower half of the Todd River Dolomite.

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The fourth sampling locality, Central Mount Stuart, is in a southern outlier of the Georgina Basin and is considered to be in platform equivalents of the Precambrian–Cambrian marine rocks of the Amadeus Basin to the S. The sediments are generally flat-lying and free from obvious foliation. Zeolites have been reported, however, from the lower units at this locality (A. J. Stewart, personal communication, 1976). The sequence is a series of silicified and occasionally cross-bedded sandstones with some interbedded, slope-forming horizons of claystone and shale. At this locality, a fossil belonging to the Mt Skinner fauna, *Hallidaya brueri*, has been found in float, and the author discovered a form *in situ* which resembles *Arumbera banski* (see Glaessner & Walter, 1975). These occurrences suggest that the beds at Central Mount Stuart are late Precambrian in age.

### 3. Sampling

On 3 sampling trips between November 1975, and July 1976, a total of 390 oriented samples spanning an exposed sequence of 500 m were collected at the Ross River locality (present lat. 23.6° S, long. 134.5° E). The sampling began in the lower ledges of the Julie member of the Pertatataka Formation and continued stratigraphically upwards through the Arumbera, Box Hole, and Allua Formations (new names of Daily (1974) for the Arumbera members I, II and III, respectively) and into the Todd River Dolomite. Oriented samples were collected as hand specimens or with a portable 1 in. diamond drill. Stratigraphic intervals between samples were measured with a Jacob staff and an attached Abney hand level.

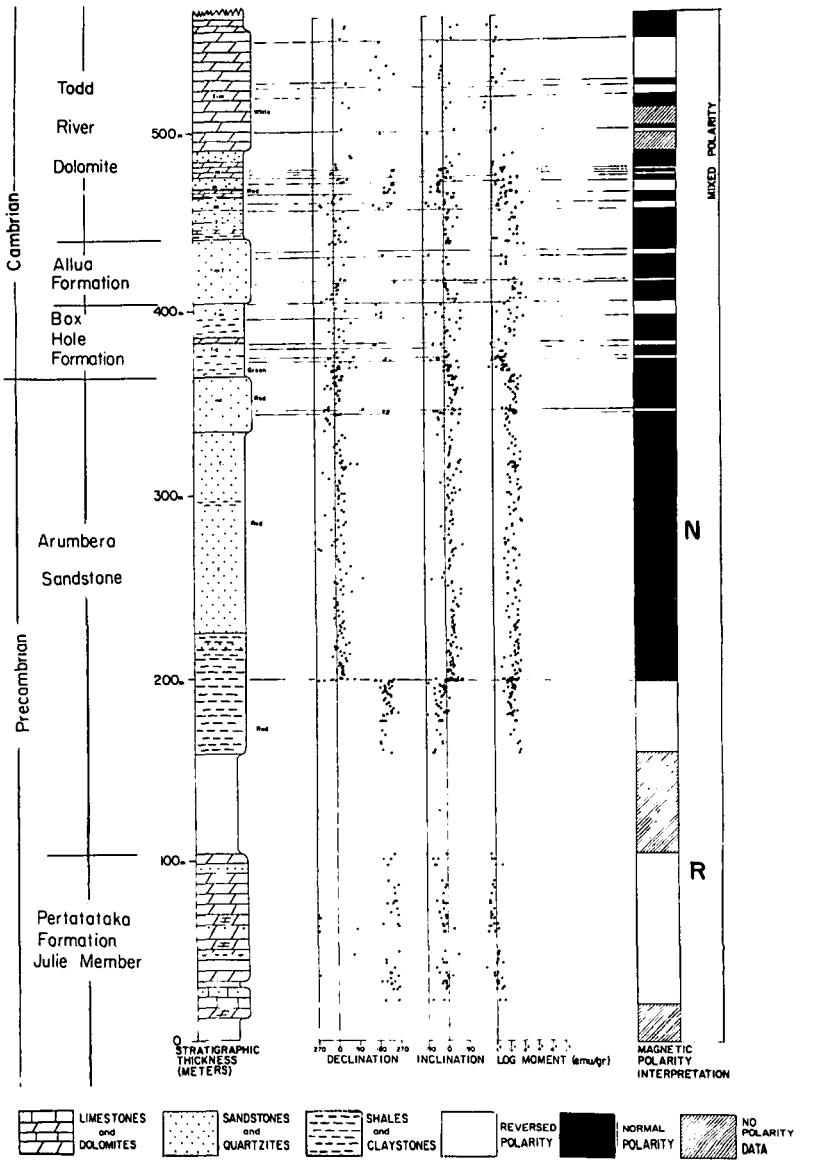
The sampling density was greatest in the red sandstones, shales, and claystones of the Arumbera Sandstone and Allua Formations, with somewhat less attention being given to the green shales of the Box Hole Formation and to the white and grey carbonates of the Todd River Dolomite and Pertatataka Formations. The stratigraphic section is almost continuously exposed along the river, with the exception of 58 m of claystone at the base of the Arumbera Sandstone. The surface samples are fresh, and only minor weathering is visible in hand samples. of the 4 localities sampled, this site is by far the best exposed.

In 2 separate excursions during April and July, 1976, a total of 220 oriented samples were collected in a similar manner from a section through the Arumbera Sandstone south of Valley Dam (present lat. 23.8° S, long. 133.6° E) near Alice Springs, Northern Territory, where Precambrian fossils have been reported (Glaessner & Walter, 1975). The sampling began about 5 m above the top of the Pertatataka Formation and continued for 410 m up through the Arumbera and the Box Hole Formations. The sampling density was highest across the fossil horizon where the exposure was good, but parts of the section above and below this were intermittently obscured; consequently, there were a number of unavoidable gaps in the sampling.

At the Areyonga locality (lat. 24.1° S, long. 132.3° E) an unfossiliferous unit, the Eninta Sandstone, mapped as an equivalent of the Arumbera, rests with marked unconformity on red and green claystones of the Pertatataka Formation. The Pertatataka at this site is stratigraphically below a carbonate unit identified as the Pertatataka's Julie member (A. T. Wells, personal communication, 1976). On stratigraphic grounds alone, it is impossible to determine the exact amount of time represented by the unconformity at this point, but at the minimum the equivalent of the Julie member of the Pertatataka was eroded from the Areyonga sequence. A total of 135 oriented palaeomagnetic samples covering 260 m of section in these formations were collected from essentially continuous exposure in the bottom of Areyonga Creek and in a nearby wash.

The Central Mount Stuart Formation at Central Mount Stuart was the least densely sampled of the 4 localities. A total of 41 oriented hand-samples were collected from 200 m of stratigraphic section which spanned an isolated occurrence of *Hallidaya brueri*, a component of the Mt Skinner fauna.

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ROSS RIVER GORGE, N.T.

Figure 2. Demagnetized palaeomagnetic directions and polarity interpretations for the Ross River locality, Central Australia.

## 4. Demagnetization data and magnetic polarity interpretations

All samples were progressively stepwise demagnetized at temperature steps varying between 100 and 680 °C in a furnace developed by McElhinny, Luck & Edwards (1971). Red samples from the Arumbera and Allua Formations were generally processed to 680 °C in between 8 and 12 steps, in contrast with the green beds and carbonates which were treated in 7–10 steps between 100 and 475 °C. Magnetic measurements were taken on an ScT superconducting rock magnetometer which had been interfaced by the author to a DIGICO mini-computer for immediate data reduction.

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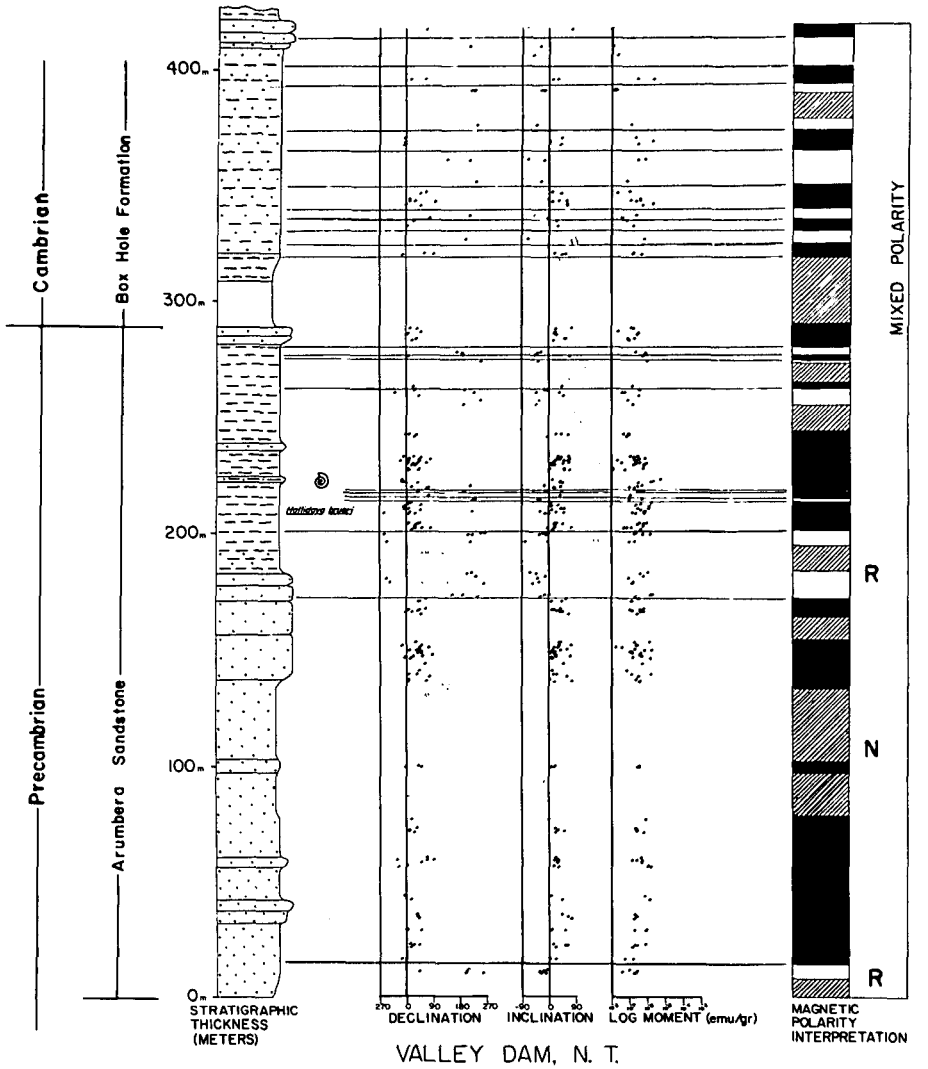
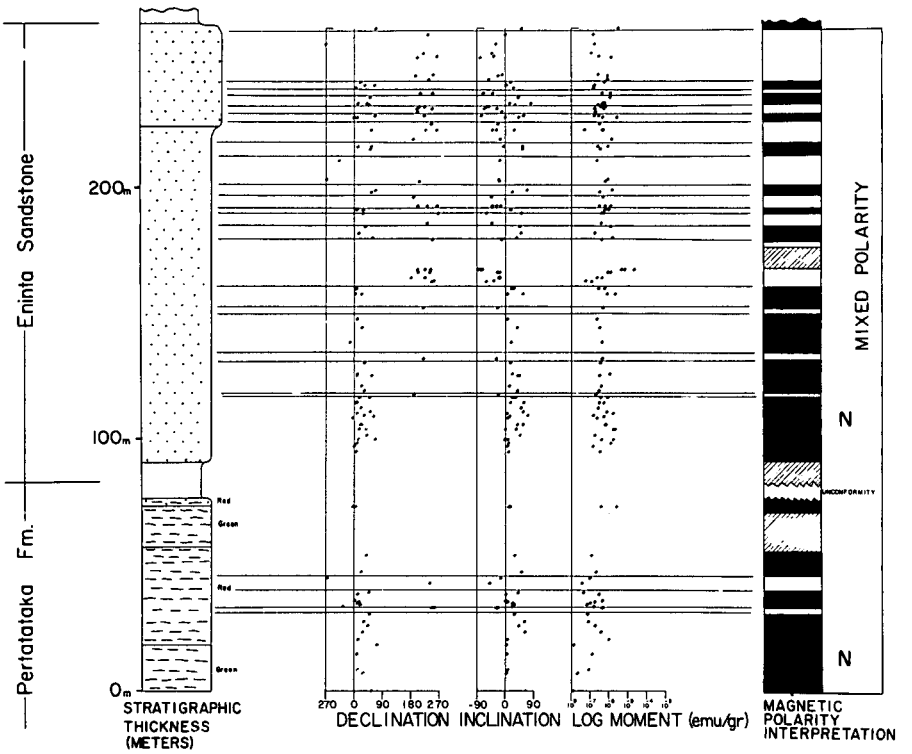



Figure 3. Demagnetized palaeomagnetic directions and polarity interpretations for the Valley Dam locality, Central Australia. Lithologic and polarity symbols are the same as on Figure 2.

Directions and intensities of specimen magnetic vectors after each thermal demagnetization step were plotted on 2 component orthogonal projections for polarity interpretation. Polarity assignments were based both on the specimen magnetic directions and demagnetization paths in a manner similar to that used by Steiner & Helsley (1975). The polarity assignments of 'normal' and 'reversed' were made by comparison with the apparent polar wander path for Australia of McElhinny & Embleton (1976). A detailed analysis of the primary and secondary components of magnetization in these rocks is in preparation (Kirschvink, 1977). Some samples, particularly the carbonates and green sediments, show considerable secondary components of magnetization which sometimes cannot be removed by thermal treatment. Polarity interpretations have therefore been based on samples which respond well to thermal demagnetization. For these specimens, the directions obtained after the highest temperature cleaning steps before unstable magnetic behaviour was observed have been plotted against their stratigraphic position in Figures 2–5. In addition, the magnetic polarity interpretations based on the total demagnetization paths for each specimen have also been shown.




AREYONGA GORGE, N. T.

Figure 4. Demagnetized palaeomagnetic directions and polarity interpretations for the Areyonga Gorge locality, Central Australia. Lithologic and polarity symbols are the same as on Figure 2.

## 5. Magnetic polarity sequences

Determination of the magnetic polarity sequence at Ross River (Fig. 2) is straightforward in most parts of the section. The Julie member of the Pertatataka Formation and the basal portion of the Arumbera Sandstone are clearly of reversed polarity. The remainder of the Arumbera is occupied by a long interval of normal polarity. An intensive re-sampling of the section in the Arumbera containing the change between the reversed and normal intervals reveals that this transition occurs within 50 cm. Samples obtained within this short interval are of intermediate direction, and may have acquired their magnetizations during the actual reversal of the geomagnetic field. Above the Arumbera-Box Hole contact, the magnetic polarity sequence appears to be made of mixed normal and reversed zones, with an initial bias toward reversed polarity gradually disappearing upwards. Although the present sampling is not sufficiently detailed to resolve the fine polarity sequence of this interval, mixed polarities certainly extend through the Todd River Dolomite.

The magnetic polarity structure of Valley Dam (Fig. 3) is basically similar to that observed at Ross River. The first 5 samples taken in the basal Arumbera Sandstone are of reversed polarity, above which is a long normal interval followed once again by a sequence of mixed polarities. The difference at Valley Dam, however, is that the mixed interval starts well below the Box Hole formation. It should be noted that the upper portion of this interval has not been sampled in great detail, and the absolute number and length of the magnetic zones cannot be determined at this time.

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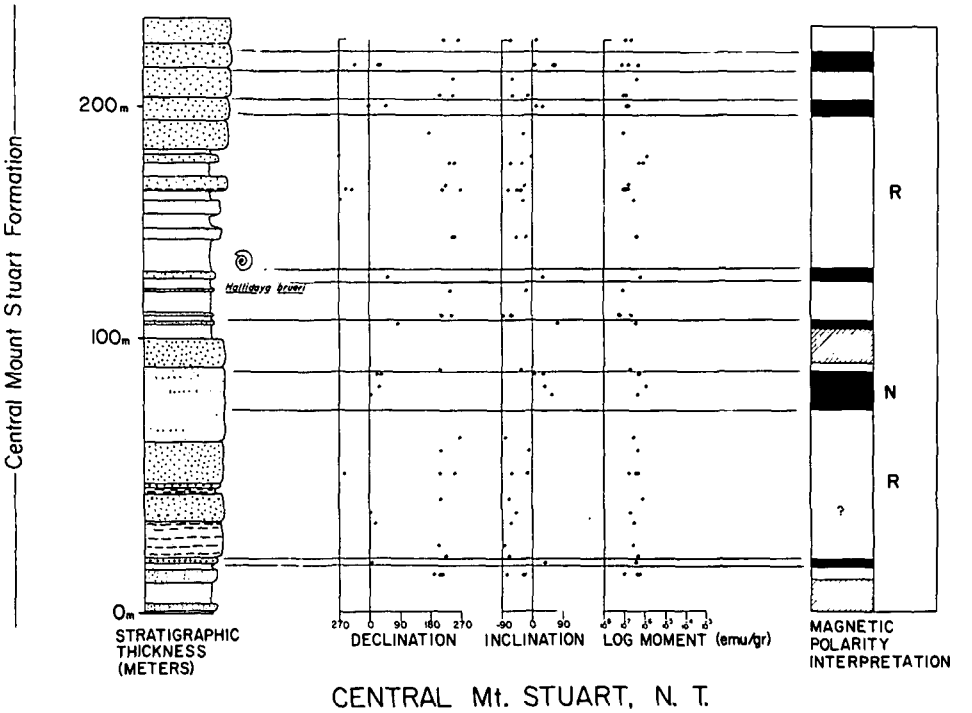



Figure 5. Demagnetized palaeomagnetic directions and polarity interpretations for the Central Mount Stuart locality, Central Australia. Lithologic and polarity symbols are the same as on Figure 2. The stratigraphic column is courtesy of the Bureau of Mineral Resources in Canberra.

Mixed magnetic polarity sequences also occur both at Areyonga and Central Mount Stuart (Figs 4, 5). The Pertatataka claystones at Areyonga are predominantly of normal polarity and contain only 2 minor reversed zones. The overlying Eninta Sandstone seems to have a normal polarity bias at the base and thereafter grades up into a more evenly distributed mixed interval. The sampling density at Central Mount Stuart is not high enough to resolve individual magnetic zones with any accuracy, but it does show a substantial bias towards reversed polarity with occasional short normal intervals.

## 6. Magnetic polarity comparisons

The comparison between the Ross River and Valley Dam sections, about 80 km apart, is relatively simple, and is shown on Figure 6. Both localities have reversed polarity intervals at the base which are followed by long intervals of normal polarity. The intervals differ only by 5 m in length (160 at Ross River compared to 155 at Valley Dam), and could easily be a result of slightly differing sedimentation rates. A problem does arise, however, which concerns their relative stratigraphic position. At Ross River, the top of the long normal interval is very close to the Arumbera-Box Hole contact. At Valley Dam, however, the top of this interval is 85 m below the contact, and well below the horizon containing components of the Mt Skinner Fauna. Although the outcrop is not continuous between Ross River and Valley Dam, the abrupt lithological change from red to green which marks the Arumbera-Box Hole boundary is a disconformity, as has been suggested by Daily (1974).The upper units of the Arumbera Sandstone which are present at Valley Dam have been eroded away at Ross River. The gap in time represented by the Arumbera-Box Hole unconformity at Ross River at least includes a portion

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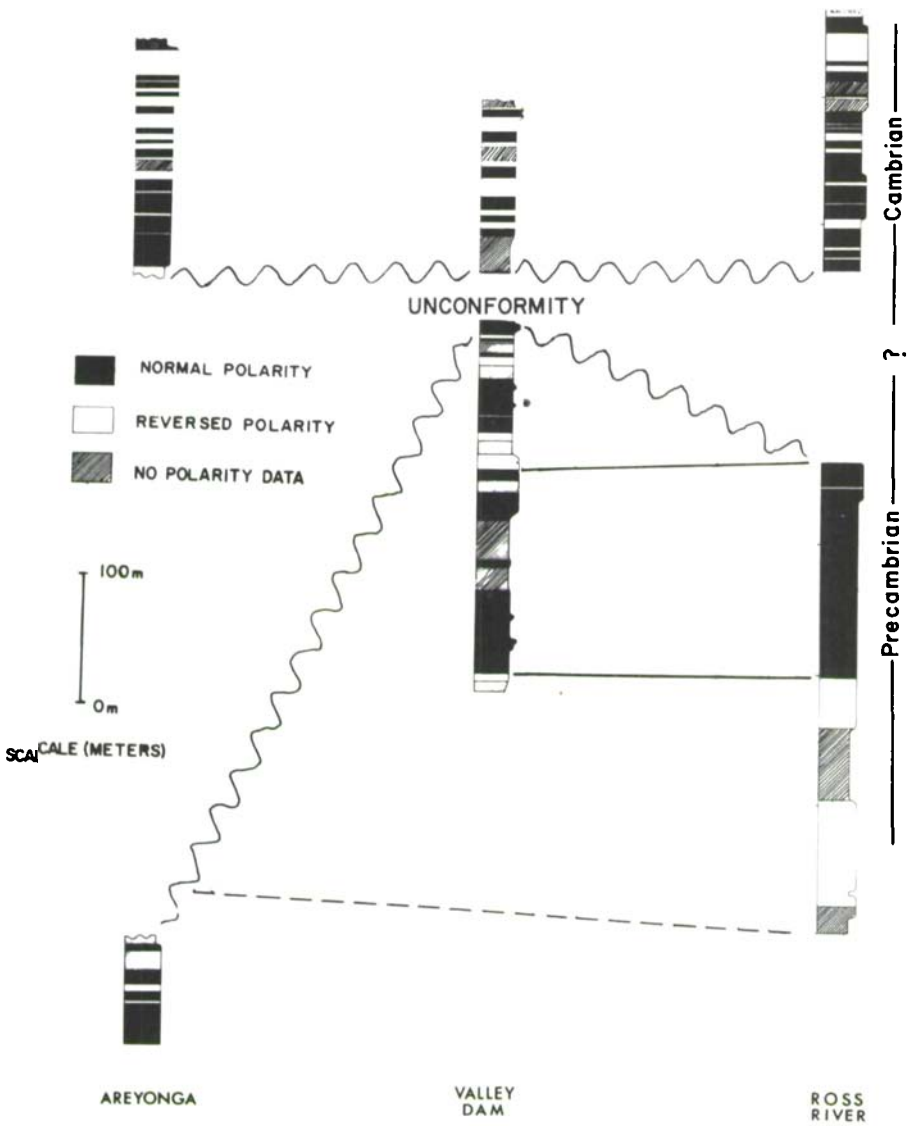



Figure 6. Palaeomagnetic correlations within the Amadeus Basin, Central Australia.

of the Mt Skinner zone, although, as will be discused later, at least 1 form of the Mt Skinner fauna is time-transgressive. It is interesting to note that the top of the long normal interval at both localities is located in a massive, ridge-forming red sandstone-quartzite which can be traced for many km in either direction. It is possible that the silicification of this unit occurred shortly after deposition. If confirmed by magnetostratigraphic studies, it may therefore be a useful time-marker for correlation within the Arumbera Sandstone. A detailed correlation of the magnetic zones in the upper half of these sections is hampered by the sparse sampling density in the top part of the Valley Dam locality. It is significant, however, that both show mixed polarity.

The magnetic comparison between Areyonga and the 2 sections discussed above is not as simple. On stratigraphic grounds, the portion of the Pertatataka Formation which was sampled at Areyonga should fit somewhere below the sampled section at Ross River. This, as discussed

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earlier, is because further to the W a carbonate identified as the Julie member (A. T. Wells, personal communication) is found above the units of the Pertatataka sampled in Areyonga creek. The predominantly normal interval in the base of the Eninta Sandstone appears to match best with the interval above the base of the Box Hole formation at Ross River, as both of these intervals are dominantly of normal polarity with minor reversed zones. It is quite possible that some very short polarity zones both at Ross River and Areyonga have been missed, and a complete, detailed comparison will require additional intensive sampling at both localities. It is also possible that the time-equivalent beds to the Eninta Sandstone hae been cut away by the unconformity at the Arumbera-Box Hole contact at Ross River. It is interesting to speculate that these unconformities which occur in different parts of the Amadeus basin might actually be the same, or lateral equivalents caused by the same tectonic activity, which may be the Petermann Ranges Orogeny. This is certainly consistent with the magnetic polarity stratigraphy.

The magnetic polarity sequence observed in the Central Mount Stuart Formation at Central Mount Stuart does not fit well with the other 3 sections. Although the sampling interval is sparse, the data (Fig. 5) resolve a reversed polarity bias with short normal zones. The only part of the sequence observed elsewhere which is biased in a reversed sense is the Julie member of the Pertatataka Formation and the lower portion of the Arumbera Sandstone at Ross River. The Central Mount Stuart sequence might well fit somewhere in this part of the polarity pattern either below the Julie or in the 58 m of claystones which are not exposed at the base of the Arumbera Sandstone. An alternative position would be in the disconformity between the Arumbera and the Box Hole Formations. In any case, the magnetic polarity pattern does not match that found at the fossil horizon in the Arumbera Sandstone at Valley Dam, which is predominantly normal with short reversed zones (Fig. 7). Both localities, however, have produced the fossil, *Hallidaya brueri* Wade, which is an element of the late Precambrian Mt Skinner fauna. If the identification of this fossil is correct at both of these localities, then *H. brueri* must have a considerable vertical distribution in time, and care should be used in interpreting the age of late Precambrian strata in which it, and it alone, is found. Better, more confident, age assignments could be made if more than 1 element of a Precambrian faunal assemblage were to be located within a stratigraphic unit.

## 7. Discussion

The Upper Proterozoic to early Cambrian rocks of Central Australia were apparently formed at a time when the Earth's magnetic field was behaving in a distinctive manner. The rocks record 2 successive, relatively quiet intervals of opposite polarity, first reversed then normal followed by an interval of high reversal frequency. The available fossil evidence from the Arumbera Sandstone shows that both these quiet intervals are Precambrian, and that the base of the Cambrian must lie within the mixed interval. At the present time, no direct relationship between the onset of this mixed interval and the evolutionary events leading to the base of the Cambrian can be ascertained; however, the fact that the base of the Cambrian is near such a disturbed interval cannot be omitted from early Cambrian palaeo-environmental studies.

This simple polarity sequence offers the possibility of bridging the gap between the biostratigraphic and radiometric geochronologies for the Precambrian–Cambrian boundary. One would expect this pattern to emerge from a properly detailed palaeomagnetic study of dated igneous bodies. However, as discussed by Irving & Pullaiah (1976), the palaeomagnetic data beyond 350 Ma is not sufficient to resolve these long polarity intervals at the present time.

It is possible for the observed intervals of low reversal frequency in the Arumbera Standstone to be a result of wildly varying sedimentation rates within the Amadeus Basin. However, as noted earlier, there is a considerable lateral variation in lithology within the basin. The Todd River Dolomite at Ross River and the Eninta Sandstone at Areyonga both have similar magnetic reversal

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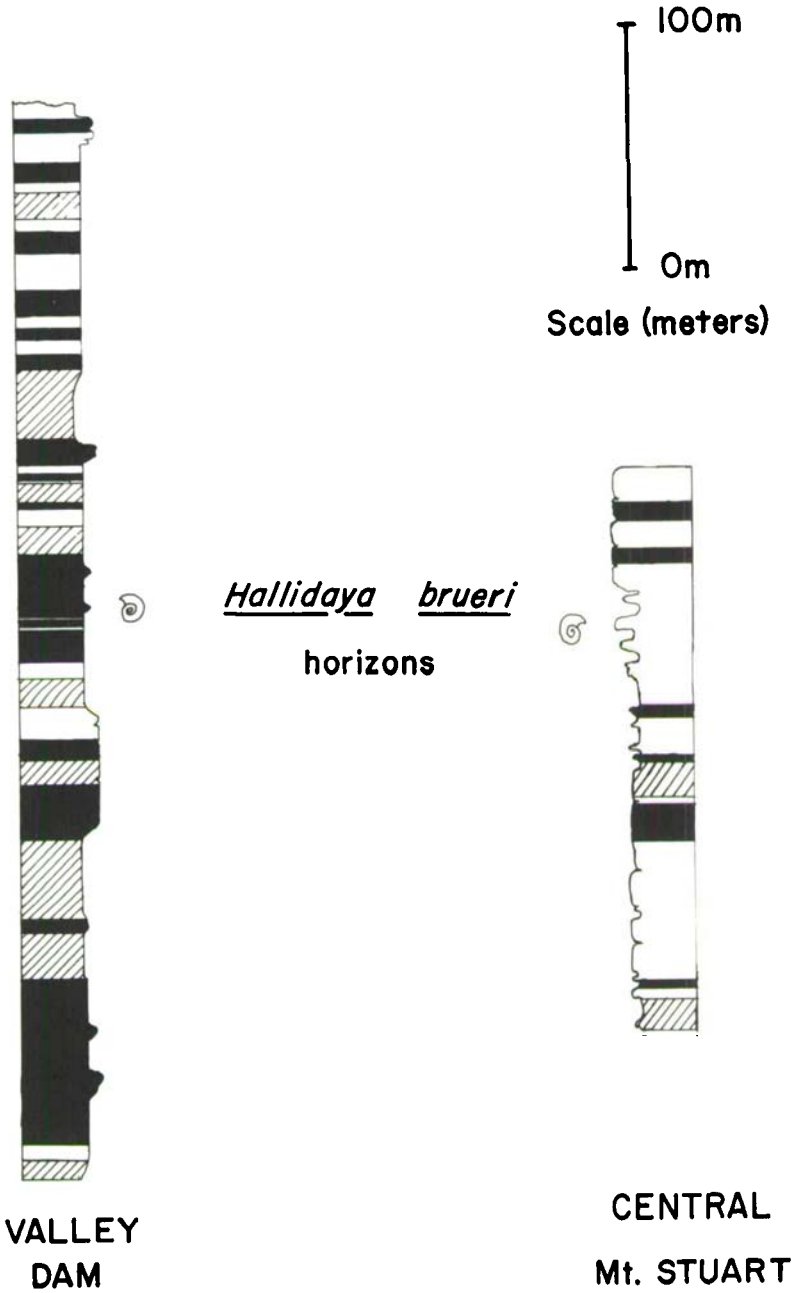



Figure 7. Magnetostratigraphic non-correlation between 2 occurrences of the late Precambrian Mt Skinner fossil *Hallidaya brueri*.

frequencies, but their lithologies are totally different. This suggests that the observed facies are more dependent on factors such as distance from source areas than on sedimentation rate. Secondly, the Eninta Sandstone at Areyonga is lithologically very similar to the lower Arumbera Sandstone at Valley Dam, yet the reversal frequencies are totally different. In view of these lithological differences, the simplest explanation for the observed polarity sequence is that it reflects an actual variation in the reversal frequency of the palaeogeomagnetic field. These quiet intervals are not uncommon in the palaeomagnetic record.

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If this polarity pattern is indeed an accurate recording of the behaviour of the geomagnetic field, then the single reversal in the basal Arumbera Sandstone is somewhat unique, in that it separates 2 thick, quiet intervals of opposite polarity. For this to occur, whatever internal polarity biasing mechanisms that exist within the Earth must switch direction along with the primary magnetic field. No reversal of this nature has previously been studied, and this one warrants further investigation. It is interesting to speculate that, unlike recent reversals, the magnetic field during this transition may have retained either a dipolar or quadrapolar form. If either were true, a detailed examination of this transition on 2 or more separate continents would yield information about their relative palaeolongitudes. The transition does appear to be present at the Ross River locality in Central Australia.

It would pose severe problems in defining just where the Precambrian–Cambrian boundary is, however, if further examination revealed the presence of Lower Cambrian fossils within the quiet interval of normal polarity. The Working Group on the Precambrian–Cambrian boundary has generally agreed that the base of the Cambrian should be placed somewhat below the earliest olenellid trilobite, but later than soft-bodied metazoans similar to the Ediacaran fauna. There is no *a priori* reason why these forms could not be partly contemporaneous; indeed, soft-bodied metazoans have persisted throughout geological time, outliving even the trilobites. In addition, very little is currently known about the geographical distribution of the early metazoans. Trilobites or other organisms evolving on 1 continental shelf could have been geographically isolated from those on another. If problems of this nature arise from future work, a strong case could be made for choosing a stratotype sequence for the Precambrian–Cambrian boundary which appears to possess nearly penecontemporaneous magnetization. The boundary could then be chosen at a point in the sequence which corresponds to the margin of a geomagnetic quiet interval. Hopefully, at the stratotype section the reversal chosen would also fulfil the international Working Group guideline cited above. A similar case has been discussed by Nairn & Peterson (1973) for the Permo-Triassic boundary.

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## References

Cloud, P. 1973. Possible stratotype sequences for the basal Palaeozoic in North America. *Am. J. Sci.* **273**, 193–206.

Cowie, J. W. & Glaessner, M. F. 1975. The Precambrian–Cambrian boundary: A Symposium. *Earth-Sci. Rev.* **11**, 209–51.

Daily, B. 1974. The Precambrian–Cambrian boundary in Australia. Specialist group in biostratigraphy and paleontology: Precision in Correlation, Programmes and Abstracts, Hobart.

Glaessner, M. F. & Walter, M. R. 1975. New Precambrian fossils from the Arumbera Sandstone, Northern Territory, Australia. *Alcheringa* **1** (1).

Irving, E. & Pullaiah, G. 1976. Reversals of the geomagnetic field, magnetostratigraphy, and relative magnitude of palaeosecular variation in the Phanerozoic. *Earth-Sci. Rev.* **12**, 35–64.

Kirschvink, J. L. 1977. The Precambrian–Cambrian Boundary Problem: Primary and secondary palaeomagnetic directions from the Amadeus Basin, Central Australia. (In preparation.)

McElhinny, M. W. & Embleton, B. J. J. 1976. Precambrian and Early Palaeozoic Palaeomagnetism in Australia. *Phil. Trans. R. Soc. Lond.* A **280**, 417–31.

CAMBRIDGE JOURNALS

McElhinny, M. W., Luck, G. R. & Edwards, D. 1971. A large volume magnetic field-free space for thermal demagnetization and other experiments in palaeomagnetism. *Pure Appl Geoph.* **90**, 126–30.

Nairn, A. E. M. & Peterson, D. N. 1973. A review of Permian and Triassic palaeomagnetic data with respect to palaeogeographic conditions at, and the location of, the Permian–Triassic boundary. In *The Permian and Triassic Systems and Their Mutual Boundary* (ed. A. Logan and L. V. Hills), pp. 694–713. *Can. Soc. Pet. Geol.* Memoir 2.

Smith, K. G. 1972. Stratigraphy of the Georgina Basin. *Bull. Bur. Min. Res.* **111**, 1972.

Steiner, M. R. & Helsley, C. E. 1975. Reversal pattern and apparent polar wander for the Late Jurassic. *Bull. geol. Soc. Am.* **86**, 537–1543.

Wells, A. T., Forman, D. J., Ranford, L. C. & Cook, P. J. 1970. Geology of the Amadeus Basin, Central Australia. *Bull. Bur. Min. Res.* **100**.

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